Julian Fletcher

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Education

University of Missouri — Columbia, Missouri

M.S. Computer Science (GPA: 4.0/4.0) — Expected May 2027

B.S. Computer Science (GPA: 3.68/4.0) — May 2026

– Minors: Mathematics, Geographic Information Systems, Information Technology

Research

Turn-Radius Constrained Pathfinding — April 2024 – Present

Undergraduate Researcher, McNair Scholar — University of Missouri

- Developed a novel framework for computing turn-radius constrained paths for autonomous vehicle navigation and kinodynamic planning.
- Created two solution paradigms: Platform-dependent (embedding constraints in graph topology) and platform-independent (online filtering for heterogeneous fleet routing).
- Developed and implemented an $O(|E| \log |V|)$ pruning mechanism using maximum-subtree-length metadata to handle 2D range queries for valid candidate edges.
- Mathematically proved all turns in a discrete path occur over a physically traversable distance $\geq L_{\min}$.
- Established a formal proof of equivalence demonstrating that both paradigms produce identical optimal solutions.
- Conducted comprehensive testing demonstrating elimination of false-positive paths in high-resolution grids.

Geospatial Engineering

SAR Flood Extent Mapping | *Sentinel-1, SNAP, ArcGIS Pro* — Oct 2024 – Dec 2024

- Mapped 2024 Minnesota River flood extents by executing a multi-stage SAR processing workflow, bypassing cloud-cover limitations inherent in optical sensors.
- Performed pre-processing in SNAP including thermal noise removal, backscatter normalization (σ^0), Gamma Map speckle filtering, and Range-Doppler Terrain Correction using Copernicus DEM.
- Conducted change detection and thresholding analysis (backscatter ≤ -17 dB) to isolate floodwater, identifying significant inundation along the river corridor despite post-crest acquisition.

Austin TX Crash Dashboard | *ArcPy, Python, ArcGIS Pro* — April 2024 – May 2024

- Developed a custom dashboard displaying Austin, TX crash data by district to identify geographic trends and spatial patterns across large incident datasets.
- Architected a data processing pipeline utilizing ArcPy to systematically ingest, join, and aggregate raw spatial data into the geographic district framework.
- Automated the underlying data visualization and formatting workflow using Python and Jupyter Notebooks.

Professional Experience

Enterprise Mobility — St. Louis, MO

Software Engineering Intern — May 2025 – July 2025

- Integrated into a cross-functional Agile team responsible for replacing two legacy systems for global brands (Enterprise, National, Alamo).
- Engineered a billing transparency feature; updated React UI components and integrated RESTful APIs to display refueling charge specifics.
- Implemented French Canadian localization (i18n) for billing modules to support international regulatory compliance.
- Participated in regular Agile meetings, including story pointing and sprint planning.
- Conducted extensive repository refactoring to resolve Material UI (MUI) import conflicts and mitigate file descriptor exhaustion on Windows development environments.

Technical Proficiencies

Programming Languages: C++, Python, JavaScript (React.js), TypeScript, SQL, C, HTML/CSS
Geospatial Technologies: Sentinel-1 SAR, ArcGIS Pro, ArcPy, SNAP, Change Detection, Spatial Indexing
Systems & Architecture: Git, CMake, RESTful APIs, System-Level Debugging, Agile (SDLC)
Algorithmic Expertise: Kinodynamic Planning, Computational Geometry, Graph Theory, Complexity Analysis